



Biochar Plant

Carbon Removal through High tech Pyrolysis

CARBON MARKET OVERVIEW

- **GLOBAL:** 2025 market snapshot from CDR shows that from 2022 to mid-2025, around 3.04 million tonnes (Mt) of biochar carbon removal (BCR) credits were contracted. The biochar market also saw strong yearly growth, especially from 2023 to 2024, with a 435% increase. However, the number of active buyers remains low, limited to a few big companies, e.g. [Microsoft](#), [Google](#), [JPMorgan Chase](#), indicative of risk perceptions within the RE / Decarbonization projects investment community.
 - On the supply side, biochar credits remain constrained. Many projects face delays in certification or in building out production capacity. Recent data shows that some biochar projects are still validating or just issuing their first credits. This limits the credits available for immediate delivery or sale.
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- **INDIA:** The theoretical potential is very high, linked to about 500 million tons Agriculture residue and potential for high yield Bamboo plantations. Furthermore, for Net Zero target, it's essential to scale up CDR (CO₂ Direct Removal) through Biochar sequestration, in addition to NBS. Additionally, the demand for decarbonizing Steel, aligned with Green Steel Taxonomy announced by Govt of India, with biochar finding application in iron ore pellets, injection into Electric Arc Furnaces as well as for replacement of imported coking coal.
 - Despite its theoretical demand of over 10 million tons, biochar production volumes remain miniscule, due to absence of industrialized approach to feedstock management and biochar production as well as high integrity MRV for carbon accounting, which undermine confidence of bulk buyers as well as investors.
 - Google agreement to purchase 100,000 tons of carbon removal credits by 2030 from Varaha and Tata Steel use of 30,000 TPA biochar to replace Met Coal remain highlights.

THE OPPORTUNITY

Waste Management Crisis

Agricultural and forestry residues are often burned or left to decay, releasing CH_4 and CO_2



Climate Urgency

The need for "Permanent" CDR (Carbon Dioxide Removal) to meet Net-Zero goals



Soil Degradation

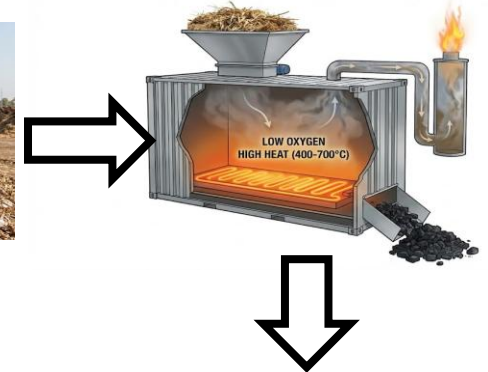
Traditional farming is depleting soil organic matter, leading to reduced yields



THE SOLUTION

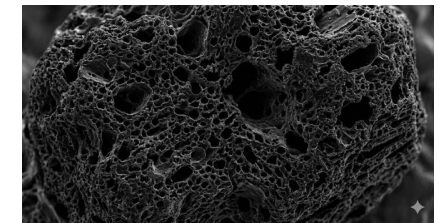
Biochar Production:

- Agricultural and forestry residues are converted into Biochar



Climate Impact:

- CO₂ Sequestration: ~2–2.5 tCO₂e per tonne of biochar → 20,000 tCO₂e annually
- Soil Benefits: enhanced soil organic carbon & water retention, yield increase & input costs reduction
- Circular Use: Biochar & compost (from farm waste) stored in agriculture fields



Triple Win Model:

- GHG mitigation
- Agriculture: Soil rejuvenation / fertility improvement & yields increase
- Livelihood: Jobs and rural entrepreneurship opportunities



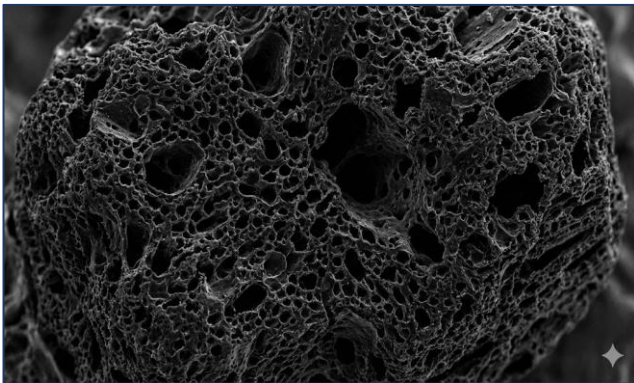
INTRODUCTION TO BIOCHAR – ENGINEERED STABILITY



Biochar is a stable, carbon-rich solid produced via pyrolysis converting unstable biogenic carbon into a stable, recalcitrant lattice. Biochar can be an important tool to increase food security and cropland diversity in areas with severely depleted soils, scarce organic resources, and inadequate water and chemical fertilizer supplies.



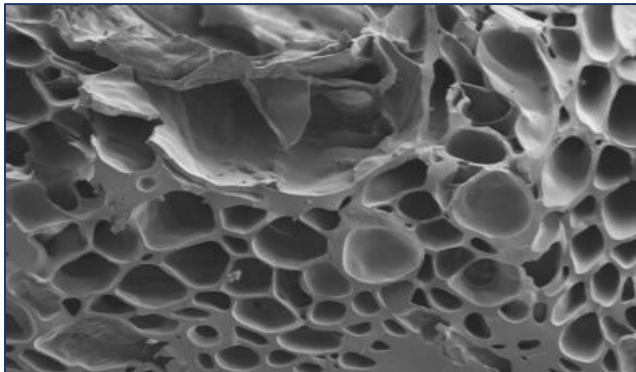
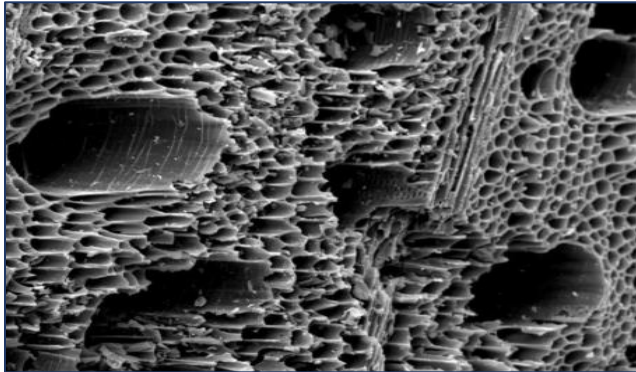
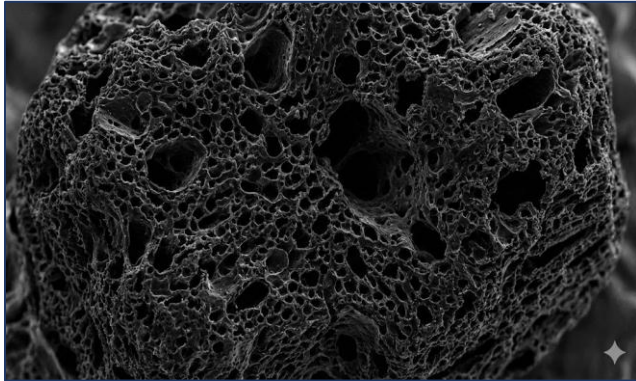
Biochar also improves water quality and quantity by increasing soil retention of nutrients and agrochemicals for plant and crop utilization. More nutrients stay in the soil instead of leaching into groundwater and causing pollution.



Biochar properties at higher temperatures

- **Higher carbon content:** Biochar produced at higher temperatures has a higher concentration of carbon because non-carbon elements are driven off as gases.
- **Increased surface area:** Higher temperatures can lead to an increase in the specific surface area and pore volume, which is beneficial for applications like adsorption.
- **Increased heating value:** The higher carbon content results in a higher heating value for the biochar.

BIOCHAR CHARACTERISTICS – THE TECHNICAL EDGE



Our Edge:

- Our Modular units with 2 nos of 1TPH feedstock intake makes it a template for quick production scaleup.

The Math:

- Every ton of biochar produced sequesters approx. **2.5–3 tons of CO₂e** (based on our **0.80 stability factor**).

Permanence:

- While biomass decomposes in 1–3 years (releasing CO₂ and CH₄), our biochar locks **carbon for 100+ years**, making it a premium product for the voluntary carbon market.

Molar Stability:

- Targeted **H:C_{org} ratio of ≤0.7**, meeting Verra VM0044 and Puro.earth requirements.

Physical Properties:

- High surface area (**>300 m²/g**) and porosity for maximum moisture and nutrient retention.

Chemical Profile:

- **High Fixed Carbon (>75%)** and Cation Exchange Capacity (CEC) to enhance soil fertility or industrial binder performance.

PRODUCTION TECHNOLOGY (THE MODULAR ADVANTAGE)

Process:

- Pyrolysis (Heating biomass in the absence of oxygen).

Efficiency:

- Automated temperature control to ensure $H:C_{org} < 0.7$, maximizing carbon credit eligibility under VM0044.

Equipment:

- Automated, continuous-feed rotary or screw pyrolysis reactors.

Autothermal System:

- The process captures volatile syngas to heat the reactor, requiring zero external fuel once started.

Precision Control:

- PLC-based systems maintain a tight temperature window (500°C – 550°C), eliminating "labile" carbon and ensuring batch consistency.



PRODUCTION PROCESS – REGISTRY ALIGNMENT

Preprocessing:

- Drying and chipping to **<15% moisture** and **<30mm size** for optimal heat transfer.

Pyrolysis:

- Oxygen-free thermal cracking in a sealed reactor.

Quenching:

- Water-cooled discharge to stop oxidation and preserve the carbon structure.

Sampling:

- Regular lab testing for H:C and O:C ratios to maintain "Carbon Gold Standard" status.

Gas Cleaning System:

- Removes tar and particulates before heat reuse.

Heat Recovery Unit:

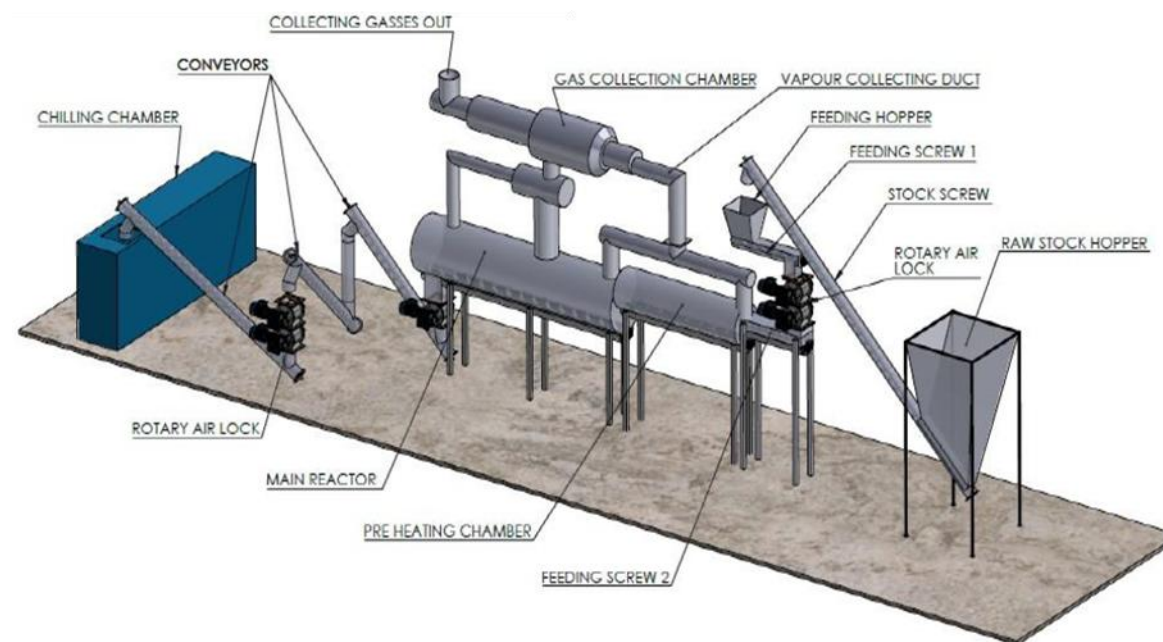
- Utilizes pyrolysis gas for drying and pre-heating.

Emission Control:

- Cyclone separator and chimney meet KSPCB norms.

Safety Controls:

- Temperature sensors, flame arrestors, and auto shutdown valves.



FEEDSTOCK SURPLUS → WASTE-TO-VALUE SUPPLY CHAIN

- **Raw Material Availability:**

- Coconut Fronds: High lignin for structural biochar; traditionally burned or left to decay.
- Paddy Straw: Addressing the regional burning crisis while producing high-silica biochar.
- Cane Trash: Tapping into massive seasonal surpluses to ensure 330 days of annual operation.

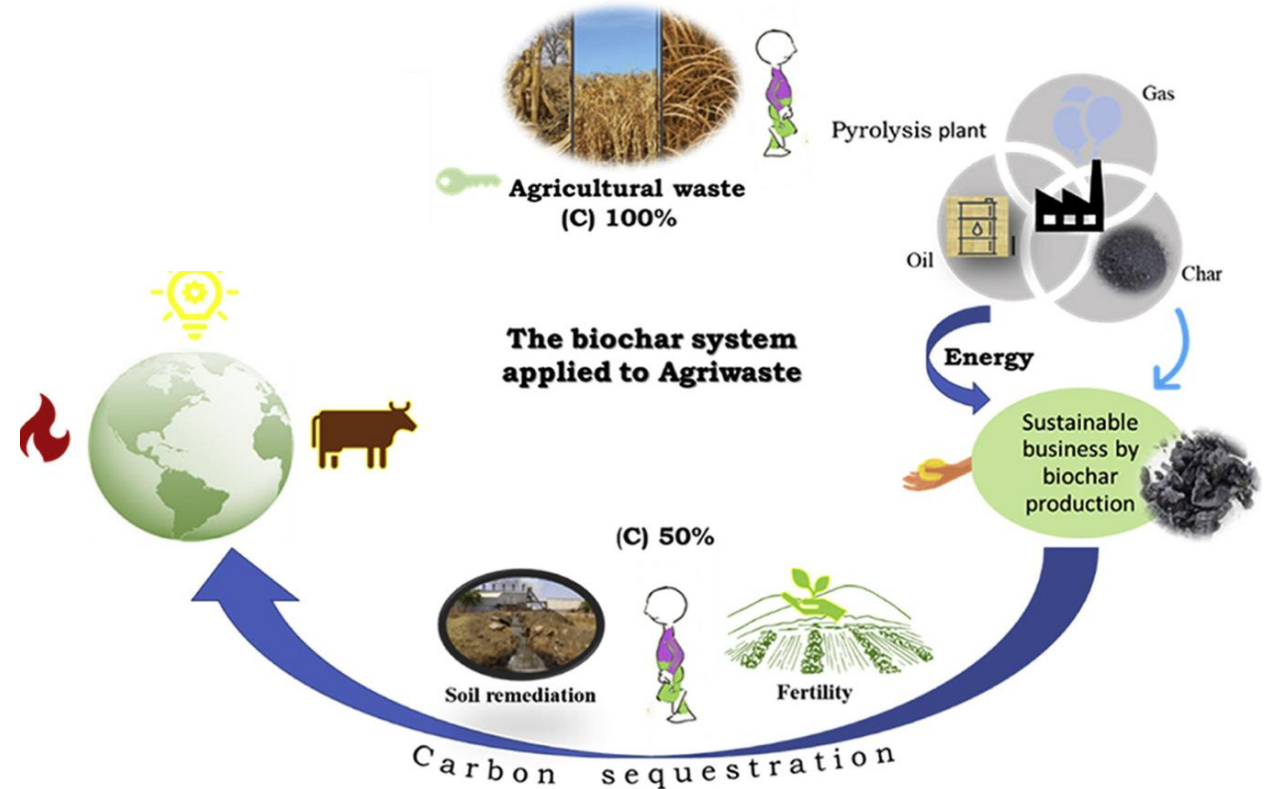
- **Sourcing:**

- Get into contract with farmers to ensure "first-right" to waste residues as the land will be cleared of agri waste, collected for pre-processing & transported to the plant



BIOCHAR FOR FARM USES (SOIL HEALTH)

- **Water Retention:**
 - Acts as a "sponge," reducing irrigation needs by up to 20%.
- **Nutrient Efficiency:**
 - Holds onto nitrogen and phosphorus, reducing fertilizer runoff.
- **Yield Increase:**
 - Proven to boost crop productivity in acidic or degraded soils.



BENEFITS OF BIOCHAR PERMANENT STORAGE IN FARMS – SUSTAINABLE AGRICULTURE + CDR

SOIL IMPROVEMENT; IMPROVES FERTILIZER EFFICIENCY; WATER RETENTION; MICROBIAL ACTIVATION; REDUCES SOIL ACIDITY; BOOSTS CROP YIELD; CARBON SEQUESTRATION



BIOCHAR FOR NON-FARM USES (INDUSTRIAL)

BIOCHAR is a carbon-neutral alternative to coal and coke, reducing CO₂ emissions and improving energy efficiency. Application in production of iron ore pellets (fed to DRI or BF Units) or as part of inputs materials to Electric Arc Furnace (EAF) and Blast Furnace (BF)

BIOCHAR ADDED TO IRON ORE WHILE PELLETIZATION, around 15%, improves pellet strength and reduction efficiency, though excessive amounts can reduce weight loss and conversion rates

BIOCHAR ADDED TO FEED MATERIALS OF ELECTRIC ARC FURNACE: acts as REDUCTANT: carbon-based reductant in the steelmaking process, absorbing oxygen and assisting in the reduction of residual FeO in DRI + SLAG FOAMING AGENT: Injecting carbon into the molten slag is crucial for creating a "foamy slag," which improves energy efficiency and protects the furnace.

BIOCHAR ADDED TO FEED MATERIALS OF BLAST FURNACE: Biochar can be injected into blast furnaces via tuyeres (air nozzles) as a partial replacement for pulverized coal (also be blended with the coke that is used in the furnace). THUS REPLACING EXPENSIVE IMPORTED METALLURGICAL COAL

Tata Steel has successfully pioneered the usage of biochar in its Jamshedpur plant. Beginning as a trial in January 2023, the Company has replaced approximately 30,000 tonnes of fossil fuel with biochar, to date. It's anticipated to reduce 50,000 tonnes of carbon dioxide emissions annually.

DIGITAL MRV – REAL-TIME TRANSPARENCY

DIGITAL M.R.V. – END- TO-END DIGITAL TRACKING

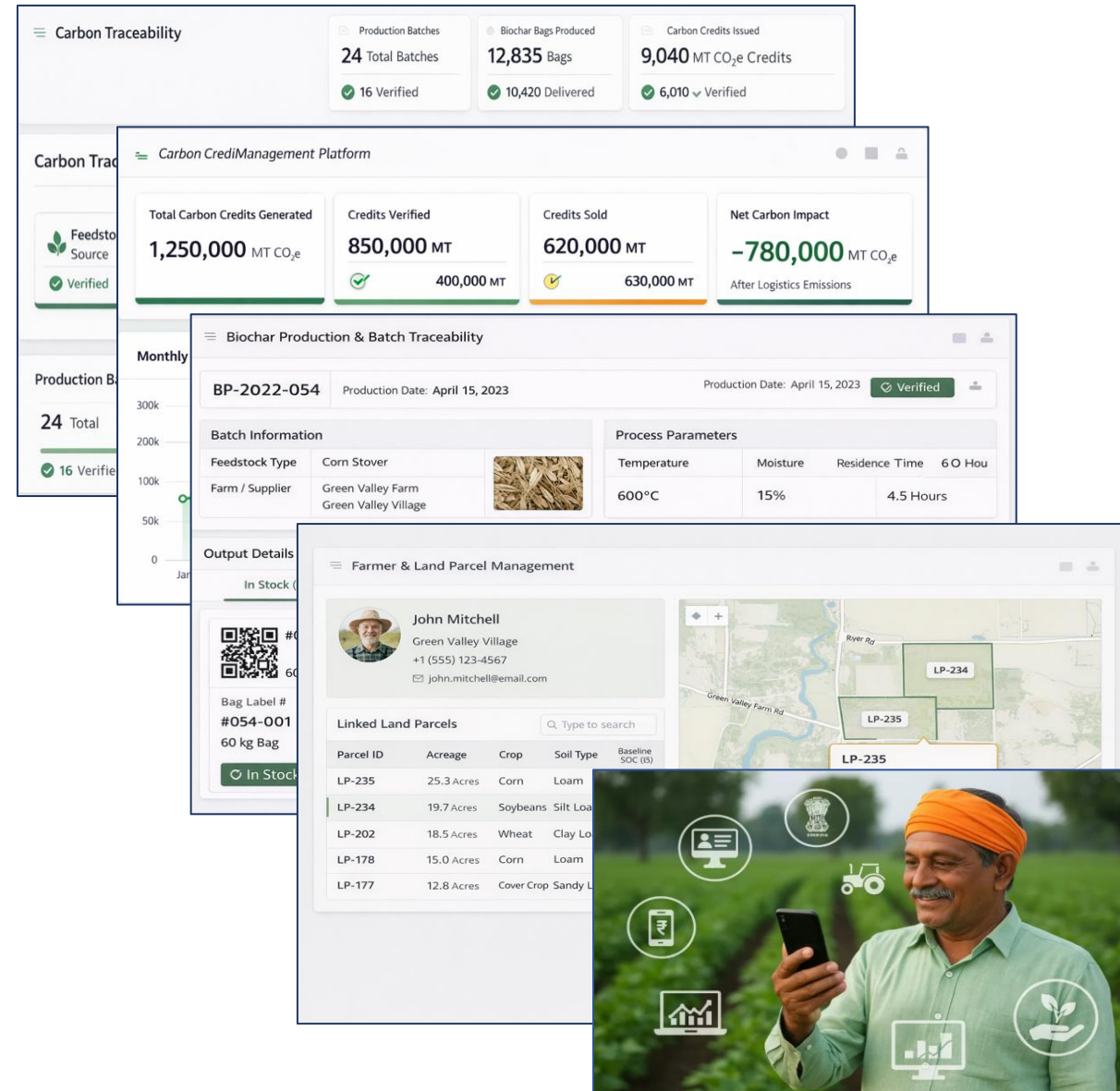
- **Biomass Collection:** Geo-tagged, timestamped at source
- **Production:** Batch tracking with RE-verified inputs
- **Soil Application:** Compost integration + field-level deployment logs

BLOCKCHAIN-ENABLED TRANSPARENCY

- Immutable data from field to credit issuance
- Smart contracts validate and time-stamp each step

REAL-TIME MONITORING PLATFORM – LIVE DASHBOARD FOR INVESTORS & BUYERS

- Performance metrics: tCO₂e removed, hectares restored, biochar pH & C-content
- Remote auditing, automated reporting for carbon registries



ENVIRONMENTAL & SOCIAL IMPACT (ESG)



UN SDGs:

- **Contributing to**
 - **SDG 1 (No Poverty)**
 - **SDG 3 (Good Health and Well-being)**
 - **SDG 4 (Quality Education)**
 - **SDG 5 (Gender Equality)**
 - **SDG 6 (Decent Work and Economic Growth)**
 - **SDG 9 (Industrial, Innovation and Infrastructure)**
 - **SDG12 (Responsible Consumption)**
 - **SDG 13 (Climate Action)**
 - **SDG 14 (Life Below Water)**
 - **SDG 15 Life on Land**
 - **SDG 17 (Partnerships for the Goals)**

COMPETITIVE LANDSCAPE (THE CDR HIERARCHY)

Feature	Reforestation (Nature-based)	Modular Biochar (Our Project)	Direct Air Capture (DAC)
Permanence	Low (30–50 years; fire risk)	High (100+ years) $F_{stable} = 0.80$	Ultra-High (1,000+ years)
Cost per tCO ₂	Low (\$15 – \$30)	Mid (\$100 – \$200)	Very High (\$600 – \$1,000)
Co-Benefits	Biodiversity	Soil Health & Waste Mgmt	None (pure removal)
Scalability	Land-constrained	Rapid (Modular/Containerized)	Capital-intensive